

ANTIDEPRESSANT DRUGS

BY

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DEPRESSION

MOOD DISORDER THAT CAUSES

- ☐ **SADNESS**
- ☐ **LOSS OF INTEREST IN ACTIVITIES**
- ☐ **SOMNOLENCE**
- ☐ **DECREASE APPETITE**
- ☐ **LOSS OF ENERGY**
- ☐ **SUICIDAL THOUGHTS**

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TYPES OF DEPRESSION

- 1. MAJOR DEPRESSION**
- 2. PERSISTENT DEPRESSIVE DISORDER.**
- 3. BIPOLAR DISORDER**
- 4. SEASONAL AFFECTIVE DISORDER (SAD)**
- 5. PERINATAL DEPRESSION**
- 6. PMDD**
- 7. ATYPICAL DEPRESSION**

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A 30-year-old woman has persistent low mood, loss of interest, poor sleep, and fatigue for the past 3 months. She is diagnosed with major depressive disorder and started on a first-line antidepressant.

A. Fluoxetine

B. Haloperidol

C. Diazepam

D. Carbamazepine

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MECHANISM OF DEPRESSION

**CHANGES IN THE LEVEL OF
NEUROTRANSMITTERS IN THE BRAIN THAT
HELP NERVE CELLS TO COMMUNICATE**

☐ **SEROTONIN**

☐ **DOPAMINE**

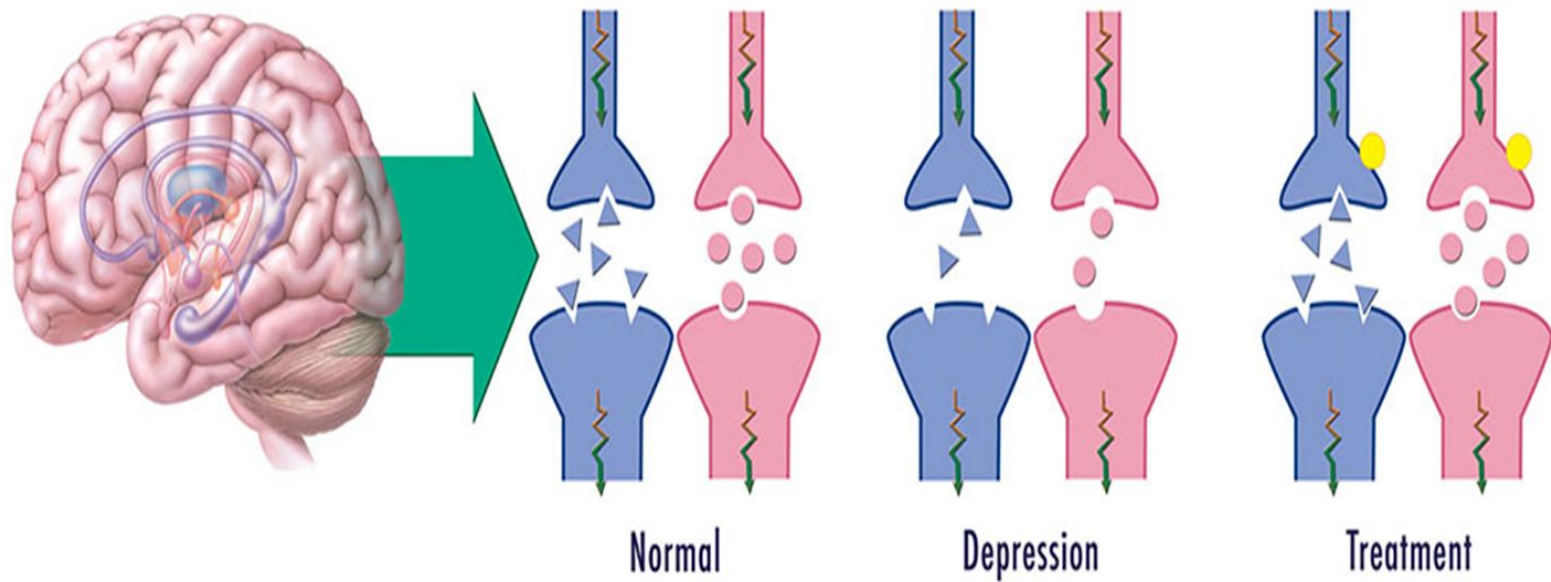
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☐ **NOREPINEPHRINE**

**THE LEVEL CAN BE INFLUENCED BY PHYSICAL
ILLNESS, GENETICS, SUBSTANCE
ABUSE, DIET, HORMONAL CHANGES, BRAIN
INJURIES OR SOCIAL CIRCUMSTANCES.**

MECHANISM OF DEPRESSION

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CLASSIFICATION OF ANTIDEPRESSANTS

1. TRICYCLIC ANTI-DEPRESSANTS (TCAS)

Tertiary amines: IMIPRAMINE , AMYTRIPTILINE, DOXEPINE

Secondary Amines: DESIPRAMINE, NORTRIPTYLINE, PROTRYPTYLINE

Tetracyclic: Amoxapine, Maprotiline

2. MONOAMINE OXIDASE INHIBITORS (MAO)

SELEGILINE, PHENELZINE , ISOCARBOXAZID, TRANYLCPROMINE

3. SELECTIVE SEROTONIN REUPTAKE INHIBITORS (SSRIS)

FLUOXETINE, CITALOPRAM, ESCITALOPRAM, FLUOXAMINE,
PAROXETINE, SERTRALINE

4. ATYPICAL ANTI-DEPRESSANTS

BUPROPION, MIRTAZAPINE, NEFAZODONE, TRAZADONE,
VORTIOXETINE

5. SEROTONINE/NOREPINEPHRINE REUPTAKE INHIBITORS (SNRIS)

DESVENLAFAXINE, DULOXETINE, LEVOMILNACIPRAM

SSRIs

- ☐ AGENTS SELECTIVELY INHIBIT SEROTONIN REUPTAKE (INHIBIT SERT)
- ☐ LITTLE BLOCKING ACTIVITY AT MUSCARINIC, ALFA ADRENERGIC AND HISTAMINE H1 RECEPTORS
- ☐ SAFEST ANTIDEPRESSANT FOR USE
- ☐ DRUG OF CHOICE IN TREATING DEPRESSION

MECHANISM OF ACTIONS

- ☐ BLOCK REUPTAKE OF SEROTONIN (INHIBIT SERT)**
- ☐ RESULTS IN INCREASED CONCENTRATIONS OF SEROTONIN IN SYNAPTIC CLEFT**
- ☐ TAKE 2 WEEKS TO PRODUCE ANTIDEPRESSANT EFFECT**
- ☐ MAXIMUM BENEFIT REQUIRE 12 WEEKS OR MORE**

CLINICAL INDICATIONS OF SSRIS

- ☐ OBSESSIVE-COMPULSIVE DISORDER
(FLUOXETINE, SERTRALINE AND
FLUVOXAMINE IN CHILDREN)**
- ☐ PANIC DISORDER**
- ☐ GENERALIZED ANXIETY DISORDER**
- ☐ POSTTRAUMATIC STRESS DISORDER**
- ☐ SOCIAL ANXIETY DISORDER**
- ☐ PREMENSTRUAL DYSPHORIC DISORDER**
- ☐ BULIMIA NERVOSA**
- ☐ CHILDHOOD DEPRESSION (FLUOXETINE
AND ESCITALOPRAM)**

PHARMACOKINETICS OF SSRI_s

- ❑ ORALLY WELL ABSORBED**
- ❑ FOOD INCREASES SERTRALINE ABSORPTION**
- ❑ HALF LIFE RANGE 16- 36 HOURS**
- ❑ METABOLISM BY CYP 450**
- ❑ FLUOXETINE HALF LIFE MUCH LONGER 50 HOURS**
- ❑ FLUOXETINE AND PAROXETINE INHIBITOR OF CYP450**

ADRS

- ☐ HEADACHE
- ☐ SWEATING
- ☐ ANXIETY
- ☐ AGITATION
- ☐ FATIGUE
- ☐ INSOMNIA (FLUOXETINE, SERTRALINE)
- ☐ SOMNOLENCE (PAROXETINE, FLUOXAMINE)
- ☐ SEXUAL DYSFUNCTION (LESS WITH BUPROPION AND MIRTAPIXINE)
- ☐ SUICIDAL THINKING IN CHILDREN
- ☐ CITALOPRAM OVERDOSE CAUSES ARRHYTHMIAS
- ☐ SEIZURES
- ☐ SEROTONIN SYNDROME
- ☐ DISCONTINUATION SYNDROME (FLUOXETINE IS AN EXCEPTION)

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A 22-year-old woman with bulimia nervosa has recurrent binge-eating episodes followed by self-induced vomiting. Pharmacotherapy is planned along with cognitive behavioral therapy.

- A. Fluoxetine
- B. Amitriptyline
- C. Phenelzine
- D. Mirtazapine

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TRICYCLIC ANTIDEPRESSANTS

- ❑ INHIBIT NOREPINEPHRINE AND SEROTONIN REUPTAKE INTO THE PRESYNAPTIC NEURON (SERT AND NET)**
- ❑ ALSO BLOCK SEROTONERGIC, ALFA ADRENERGIC, HISTAMINIC AND MUSCARINIC RECEPTORS**
- ❑ AMOXAPINE ALSO BLOCKS 5-HT2 AND D2 RECEPTORS**
- ❑ THEY ELEVATE MOOD, IMPROVE MENTAL ALERTNESS, INCREASE PHYSICAL ACTIVITY AND REDUCE MORBID PREOCCUPATION IN 50% TO 70% OF INDIVIDUALS WITH MAJOR DEPRESSION**
- ❑ PHYSICAL AND PSYCHOLOGICAL DEPENDENCE IS RARE**
- ❑ CHOLINERGIC REBOUND AND DISCONTINUATION SYNDROME MINIMIZED BY SLOW WITHDRAWAL OF DRUG**

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CLINICAL USES OF TCA

- 1. MODERATE TO SEVERE DEPRESSION**
- 2. PANIC DISORDER**
- 3. BED-WETTING IN CHILDREN OLDER THAN 6 YEARS (IMIPRAMINE)**
- 4. MIGRAINE PROPHYLAXIS (AMITRIPTYLINE)**
- 5. NEUROPATHIC PAIN (AMITRIPTYLINE)**
- 6. INSOMNIA (DOXEPIN)**

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ADRS TCA

❑ ANTIMUSCARINIC SIDE EFFECTS

BLURRED VISION, XEROSTOMIA, URINARY RETENSION, TACHYCARDIA, CONSTIPATION, AGGRAVATE ANGLE CLOSURE GLAUCOMA

❑ CVS

ARRYTHMIAS

❑ ALFA BLOCKING ADVERSE EFFECTS

ORTHOSTATIC HYPOTENSION (IMIPRAMINE MOST LIKELY AND NORTRIPTYLINE LEAST LIKELY), REFLEX TACHYCARDIA, DIZZINESS

❑ H1 BLOCKING SIDE EFFECT

SEDATION

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❑ SEXUAL DYSFUNCTION LESS COMMON

❑ WEIGHT GAIN

❑ NARROW THERAPEUTIC INDEX

SEROTONIN/NOREPINEPHRINE REUPTAKE INHIBITORS (SNRI)

- ❑ VENLAFAXINE, DESVENLAFAXINE AND DULOXETINE INHIBIT THE REUPTAKE OF BOTH SEROTONIN AND NOREPINEPHRINE**
- ❑ EFFECTIVE IN TREATING DEPRESSION IN PATIENTS IN WHOM SSRIS ARE INEFFECTIVE**
- ❑ ALSO EFFECTIVE IN CHRONIC PAINFUL SYMPTOMS ASSOCIATED WITH DEPRESSION**
- ❑ EFFECTIVE IN RELIEVING PAIN ASSOCIATED WITH DIABETIC PERIPHERAL NEUROPATHY, POSTHERPETIC NEURAGIA, FIBROMYLGIA AND LOW BACK PAIN**
- ❑ HAVE LITTLE ACTIVITY AT ALFA ADRENERGIC, MUSCARINIC OR HISTAMINE RECEPTORS**
- ❑ CAN PRECIPITATE A DISCONTUINATION SYNDROME**

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SNRIs

VENLAFAXINE

- AT LOW DOSE: INHIBIT SEROTONIN REUPTAKE
- AT MEDIUM TO HIGH DOSE : INHIBIT REUPTAKE OF SEROTONIN AND NE

DULOXETINE :

- AT ALL DOSES INHIBIT SEROTONIN AND NE REUPTAKE
- MODERATE INHIBITOR OF CYP2D6

ADRS:

HEADACHE, DIZZINESS, SEXUAL DYSFUNCTION, INSOMNIA, SEDATION, CONSTIPATION, HYPERTENSION (high dose), TACHYCARDIA (high dose)

A 38-year-old woman with major depressive disorder and chronic neuropathic pain is prescribed duloxetine. The physician explains that the drug acts by increasing two neurotransmitters in the synaptic cleft.

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- A. Inhibits reuptake of serotonin and norepinephrine
- B. Blocks dopamine receptors
- C. Inhibits monoamine oxidase enzyme
- D. Blocks muscarinic receptors

A 40-year-old man has treatment-resistant depression and has failed to respond to several SSRIs and TCAs. His psychiatrist decides to prescribe an antidepressant that inhibits monoamine oxidase.

- A. Phenelzine
- B. Sertraline
- C. Venlafaxine
- D. Bupropion

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ATYPICAL ANTIDEPRESSANTS

BUPROPION

- ◆ **WEAK DOPAMINE AND NOREPINEPHRINE REUPTAKE INHIBITOR**

USES

- ◆ **DEPRESSION**
- ◆ **NICOTINE WITHDRAWAL SYNDROME**

SIDE EFFECTS

- ◆ **DRY MOUTH**
- ◆ **SWEATING**
- ◆ **TREMOR**
- ◆ **SEIZURES**
- ◆ **LOW INCIDENCE OF SEXUAL DYSFUNCTION**
- ◆ **LOW RISK OF DDI**

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CONTRAINDICATIONS

SEIZURES AND BULIMIA

MIRTAZAPINE

- ❑ ENHANCES SEROTONIN AND NOREPINEPHRINE NEUROTRANSMISSION BY ACTING AS AN ANTAGONIST AT PRESYNAPTIC ALFA 2 RECEPTORS**
- ❑ ANTAGONIST AT 5HT2 RECEPTORS**
- ❑ ANTIHISTAMINE EFFECT ----SEDATION**
- ❑ NO ANTIMUSCARINIC EFFECT**

ADVERSE EFFECTS

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- ◆ INCREASED APPETITE**
- ◆ WEIGHT GAIN**
- ❑ MARKEDLY SEDATING EFFECTIVE IN DEPRESSED PATIENTS WITH INSOMNIA**

NEFAZODONE/TRAZODONE

- ❑ WEAK INHIBITOR OF SEROTONIN REUPTAKE**
- ❑ BLOCK POSTSYNAPTIC 5-HT_{2A} RECEPTORS**
- ❑ POTENT ANTIHISTAMINE EFFECT---SEDATION**
- ❑ TRAZODONE USES IN INSOMNIA**

ADRS

- ❑ ORTHOSTATIC HYPOTENSION**
- ❑ TRAZODONE-----PRIAPISM**
- ❑ NEFAZODONE—HEPATOTOXICITY**

MONOAMINE OXIDASE INHIBITORS

LAST LINE ANTIDEPRESSANTS

- ❑ PHENELZINE, ISOCARBOXAZID, SELEGILINE**
- ❑ IRREVERSIBLY OR REVERSIBLY INACTIVATE THE ENZYME, RESULTS INCREASE STORES OF NOREPINEPHRINE, SEROTONIN AND DOPAMINE WITHIN THE NEURONS INCREASES THERE CONCENTRATION IN SYNAPTIC SPACES**
- ❑ INHIBIT MAO IN LIVER AND BRAIN**
- ❑ DUE TO FOOD DRUG INTERACTION AND DDI ---THEIR USE IS LIMITED**
- ❑ AS TYRAMINE (TOXIC SUBSTANCE) UNABLE TO DEGRADE AND RELEASE LARGE AMOUNT OF CATECHOLAMINES CAUSES HYPERTENSIVE CRISIS**

- ☐ **SELEGILINE ONLY ANTIDEPRESSANT AVAILABLE IN A TRANSDERMAL DELIVERY SYSTEM**
- ☐ **LESS INHIBITION OF GUT AND HEPATIC MAO**
- ☐ **ALSO USED IN PARKINSONISM**

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USES OF MAOIS

- ☐ **THOSE DEPRESSED PATIENTS WHO ARE NOT RESPONSIVE TO TAS AND SSRIS**
- ☐ **THOSE EXPERIENCE STRONG ANXIETY**
- ☐ **ATYPICAL DEPRESSION**

ADRS MAOI

- ☐ **FOOD-DRUG INTERACTION (HYPERTENSIVE CRISIS)**
- ☐ **ORTHOSTATIC HYPOTENSION**
- ☐ **BLURRED VISION**
- ☐ **DRY MOUTH**
- ☐ **CONSTIPATION**
- ☐ **IN COMBINATION WITH SSRIS AND MAOI-----SEROTONIN SYNDROME**

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BRAIN WARM UP QUIZ

Q1. Select secondary amines Tricyclic antidepressant?

Q2. Select SSRIs used in childhood depression?

Q3. Select SSRIs that causes arrhythmias?

Q4. Select mechanism of action of imipramine?

Q5. Select drug with potent sedating effect work by blocking 5-HT_{2a} receptor and can cause priapism?

Q6. Select the mechanism of action of Mirtazapine?

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Q7. Select clinical indication of bupropion?

Q8. Select Drug That Interact With Tyramine containing food?

Q9. Select the drug which is also effective in diabetic peripheral neuropathy?

Q10. Select the drug used in migraine prophylaxis?

OPTIONS

A)CITALOPRAM

B)VENLAFAXINE

C)NORTRIPTYLINE

D)PHENELZINE

E)AMITRIPTYLINE

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F)TRAZODONE

G)BLOCK CENTRAL PRESYNAPTIC ALFA-2 RECEPTORS AND ALSO BLOCK 5-HT2 RECEPTORS

H)FLUOXETINE

I)BLOCK SERT AND NERT

J)NICOTINE WITHDRAWAL SYNDROME

BRAIN WARM UP QUIZ

(ANTIDEPRESSANTS)

Write The Correct Option (Only Alphabet) infront The Question Number

DATE:

ROLL NUMBER:					NAME:				
Q1		Q2		Q3		Q4		Q5	
Q6		Q7		Q8		Q9		Q10	

BRAIN WARM UP QUIZ

(ANTIDEPRESSANTS)

Write The Correct Option (Only Alphabet) Infront The Question Number.

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Q1	Q2	Q3	Q4	Q5
Q6	Q7	Q8	Q9	Q10